Maria Insa

DATA SCIENTIST & VISUALISATION ENGINEER | AI DECISION-SUPPORT SYSTEMS
Edinburgh, UK | +44 7849807584 | insaiglesias.maria@gmail.com | LinkedIn | Website/Portfolio

• A Data Scientist & Visualisation Engineer with 7+ years of experience transforming complex operational and time-series data into analytical reporting, operational insights, and decision-support tools. Experienced building analytical data pipelines, validation workflows, dashboards, monitoring systems, and data-driven applications using Python, SQL, R, and cloud-based platforms.
• Worked across engineering, product, research, and stakeholder-facing environments to translate business and operational challenges into actionable analytical outputs, with a strong focus on data quality, usability, communication, and informed decision-making.

Core Expertise

• Data Analytics & Data Science: KPI reporting, operational analytics, statistical analysis, time-series analysis, anomaly detection, trend analysis, exploratory analysis, actionable insights
• Data Visualisation & Reporting: Power BI, Tableau, Grafana, Plotly, interactive dashboards, self-service reporting, data storytelling, visualisation standards and best practices
• Data Engineering & Cloud Platforms: Azure Data Explorer (ADX), Azure Data Lake, ETL pipelines, structured and analysis-ready datasets, cloud-based data workflows, data validation frameworks
• Programming & Tools: Python, SQL, R, APIs, Git, reproducible workflows, analytical tooling
• Business & Analytical Problem Solving: Translating business requirements into analytical solutions, reporting logic, root-cause investigations, operational monitoring, decision-support systems
• Communication & Collaboration: Stakeholder engagement, requirements gathering, presenting complex analysis clearly, cross-functional collaboration

Relevant Work Experience

Data Visualisation Consultant | Freelance
Remote, UK/Spain | March 2026 - Present
• Built small-scale analytical tools, dashboards, and exploratory projects focused on decision-support, and interactive visualisation
• Developed portfolio case studies and interactive web-based data visualisation projects, exploring AI-assisted development workflows, rapid prototyping, and modern frontend tooling
Technologies: Python, SQL, APIs, React, Codex, Claude

Data Visualisation & Analytics Engineer | Intelligent Growth Solutions Ltd
Edinburgh, UK | Feb 2023 - Dec 2025 | Project: "Crop Height Tracker and other"
• Analysed large-scale operational and time-series datasets [Light command, water commands, telemetry, logs] to identify trends, support operational improvements, and inform data-driven decision-making
• Built and maintained analytical data pipelines, validation frameworks, and automated reporting workflows, improving data quality, reliability, and traceability [Pre-analysis data validation, Automated HVAC report]
• Developed analytical tools [Crop Height Tracker, Light Command Visualiser, CRC Logbook] for anomaly detection, root-cause investigation, operational failure analysis, and system performance monitoring

• Designed and delivered operational dashboards and analytical applications [Grafana time-out dashboards] used to monitor KPIs, investigate failures, explore system behaviour, and support operational decision-making
• Translated business and operational challenges into analytical outputs, and reporting solutions in collaboration with engineering, product, and scientific teams [HR, CS, among other departments]
• Produced a Python visualisation package [igs-visualisation-tools] and established data visualisation standards, documentation, and catalogues to improve consistency across tools, apps and reporting
Technologies: Python, SQL, Azure, ADX, Plotly.js, Angular (eco design system), Grafana, Tableau, Power BI, Gremlin-visualiser, R, Windows/macOS, APIs, Jira, Codex

Technology & Research Fellow | Glasgow Caledonian University & Centre for Theology and Inquiry
Remote, UK/USA | Feb 2021 - Apr 2022 | Project: "Videogame for neurodivergent people"
• Led a £300k grant-funded project, developing a data-informed platform for end users from concept-delivery
• Designed and evaluated data-driven interaction systems, combining quantitative and qualitative analysis to evaluate user behaviour and inform system design decisions
• Worked directly with stakeholders and end users to gather requirements, analyse insights, and translate findings into structured outputs guiding platform development
Technologies: AWS, Java, Minecraft, Figma, Git

UX Designer (Data Visualisation) | Lupovis
Remote, UK | Oct 2020 - Feb 2021
• Designed dashboards and visualisation concepts in collaboration with clients to support cybersecurity analytics, monitoring, and operational decision-making
• Translated complex datasets and stakeholder requirements into clear interface designs for system monitoring and decision support
• Contributed to MVP delivery within a cross-functional environment, supporting funding outcomes

UX Designer (Data & Analytics Interfaces) | Glasgow Caledonian University
Remote, UK | May 2020 - Sep 2020 | Project: "3D Virtual tour of GCU - Glasgow Campus"
• Delivered a web-based platform used by 20K+ users, working with stakeholders to translate user insights and organisational needs into a functional product
• Conducted user research and iterative analysis to translate stakeholder and user requirements into structured product outputs
• Collaborated with cross-functional teams to align outputs with user and organisational requirement
Technologies: HTML, CSS, JavaScript, Angular, Git

Product Designer (Data Visualisation) | Geckotech Solutions Ltd
Edinburgh, UK | Jun 2017 - Aug 2019 | Project: "Immersive system for structural inspection"
• Contributed to rail infrastructure inspection and condition monitoring tools combining computer vision, visual analytics, and engineering datasets for assessment and maintenance planning [Innovate UK Best KTP]
• Developed interactive 2D/3D visualisation workflows to support defect detection, classification, severity assessment, and validation of AI-generated outputs within a spatial context [3D visual inspection platform]
• Delivered an optical-thermal imaging acquisition system and data pipeline to capture consistent inspection data and generate 3D infrastructure models for structural analysis and inspection workflows [Imaging system]
Technologies: C++, UE4, Blender, Linux, Python, D3, Potree, R, streamlit

Education

PhD, Human-Computer Interaction, Data Vis & AI - Glasgow Caledonian University, UK (2018-2023)
BEng (Hons), Product Design & Development - Polytechnic University of Catalonia, Spain (2013-2017)